

Haoran Feng

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EDUCATION

CUHK-Shenzhen | M.S., Integrated Circuits and Systems

Sep 2025 - Jun 2027 (Expected)

Shandong University | B.Eng., Electronic Science / Computer Science

Sep 2021 - Jun 2025

Honors: Academic Scholarship (Top 20%); Special Talent Scholarship in Innovation, Shandong University.

INTERNSHIP EXPERIENCE



Ubiquant Investment - IQuest Research | LLM & HPC Intern

Apr 2026 - Present

IQuest-Q1 (320B-A15B) MoE model development and optimization: Contributed to training, inference, and RL-rollout infrastructure for the now-released model, focusing on kernel performance, training-inference consistency, runtime reliability, and multi-GPU validation.

Training Kernel Optimization

- **FA3 deterministic SWA backward:** Investigated long-sequence backward slowdowns with Nsys/NCU, isolated the dQ semaphore chain, and implemented contributor-relative tickets aligned with the reverse scheduler. Reduced representative packed backward from 6.755 to 1.699 ms (3.98x), achieved 1.59-5.37x at 2K-12K, and delivered a wheel after 3,601 regressions and 1,000 bitwise repeat tests.
- **Sink FC1 compute/communication contention:** Built a 4xH200 dual-stream proxy to study FC1 degradation under concurrent communication. Swept NCCL CTA and DeepGEMM SM budgets across 121 coarse and 89 fine configurations. Reduced AllGatherV proxy joint span by 12.80%, including 7.58% independent gain from compute throttling at fixed communication settings.

Inference Kernels and Runtime

- **Deterministic MoE Router GEMM:** Addressed small-M decode waste under CUDA Graph constraints by integrating DeepGEMM, Triton Full-K, and persistent backends in vLLM, with shape-aware selection, graph preflight, and cache/fallback. Cut median latency of 48 router GEMMs by 73.39%; passed 135 kernel and 20 model bitwise cases. With async OE and prefix-cache hits, request throughput improved 3.81%/4.36% on TP1/TP2.
- **Asynchronous OE state:** Moved token-history and oe_input_ids construction to GPU with Triton fused hashing, removing synchronization overhead and fixing cross-step recovery/slot-reuse semantics. TP1 passed 28/28 cases with +4.99% throughput; TP2/TP4 safe paths passed 84/84 with zero mismatches and +4.6%/+3.0% decode throughput.

RL Consistency and Rollout Reliability

- **R3 Router Replay:** Connected vLLM route capture/transport to Megatron replay with response-mask alignment and mismatch/f_tau_2/KL monitoring. In an 8xH200 Qwen3-30B-A3B 20-step comparison, executed-route mismatch fell from 17-19% to zero, with f_tau_2/KL reduced 36-145x/4-7x. Extended to a 128-GPU, 200-step rollout during development of the 300B-class model.
- **FlashInfer CUDA Graph hang:** Built a two-H200 TP=2 reproducer and isolated FTZ misclassification of a Lamport sentinel using eager/graph and fused/unfused controls. Backported the upstream bitwise fix; repeated model-level graph replays showed no hang or timeout.



摩尔线程

Moore Threads | Operator & Compiler Optimization Intern

Jan 2026 - Apr 2026

TensorFlow backend engineering for MUSA GPUs: Developed TensorFlow MUSA Extension operators, optimizer support, GELU graph fusion, profiling tools, and long-run reliability fixes.

- **Operators and fusion:** Integrated muDNN GELU and repaired fusion, enabling all 11 model GELUs to fuse and reducing representative-shape latency by 36.6%. Cut Logical_Or from 21.2 to 10.7 us; implemented or repaired Adam/AdaMax/GradientDescent paths and tests.
- **Reliability:** Traced random inference crashes to shape tensors entering the device path, reworked HostMemory, and repaired resource updates/session shutdown. Improved success from about 30% at 500 iterations to 100% at 1,000, with 40K/400K/800K stress runs passing.

OPEN-SOURCE CONTRIBUTIONS

- **Mooncake - 7 merged PRs:** Fixed persistence ordering, RDMA recovery, memory-unregister lifecycle, queue failure propagation, replica selection, tenant-aware cleanup, and macOS build compatibility. [Merged PRs](#)
- **Moore Threads TME - 10 merged PRs:** Contributed GELU fusion, optimizer operators, logical-operator/profiling improvements, resource-variable fixes, and long-run reliability. [Merged PRs](#)
- **Other projects - 3 merged PRs:** [Ray #66502](#): Serve gRPC handler routing; [FlashInfer #5171](#): installed BF16 rank-major session tests; [Mirage #755](#): Qwen3 demo lm_head padding that produced invalid tokens.

PROJECTS AND RESEARCH

High-Performance Quantized LLM Runtime with PyTorch Extensions

Dec 2025 - Present

- **Deployment objective and integration:** To serve a 32B quantized model on two H200 GPUs, implemented W4A16/W8A8/FP8 backend adaptation and integrated FlashAttention-2/4, GQA, softcap, and SDPA paths into the inference runtime.
- **Correctness and tensor parallelism:** Used Compute Sanitizer to locate a gemv_kernel_g128 tail-group out-of-bounds access and added boundary guards. Implemented packed-AWQ TP=2 column/row sharding with NCCL all-reduce. All four sanitizer modes passed representative shapes with zero errors/hazards; validated numerical outputs, cross-rank tokens, and interactive generation.
- **Results:** Qwen2.5-Coder-32B W4A16 reduced checkpoint size by 70.5% and peak GPU memory by 66.8%, while increasing end-to-end output throughput by 76.8% in the evaluated two-H200 deployment.

Key-Token-Weighted CoT Distillation | Submitted to AAAI 2027

Jan 2026 - Present

- **Problem and responsibility:** Addressed uniform supervision of reasoning tokens in conventional CoT distillation. Contributed to a three-stage framework of structure recovery, key-token-weighted supervision, and preference optimization; responsible for model training, hyperparameter tuning, and vLLM TP=4 evaluation.
- **Method:** Estimated token importance from the decrease in reference-answer likelihood after perturbing individual tokens in teacher rationales. Applied importance weights to SFT and an auxiliary loss, then completed LoRA/DPO training and comparative evaluation.
- **Results:** Qwen2.5-7B-Instruct achieved 94.01%/94.00% accuracy on GSM8K/SVAMP, improving on the strongest baselines by 5.51/10.10 percentage points. Manuscript submitted to AAAI 2027.

SKILLS

- **Development and profiling:** C/C++, Python, CUDA, Triton, PyTorch Extensions; Nsight Systems, Nsight Compute, and Compute Sanitizer for kernel profiling, determinism checks, and multi-GPU regression testing.
- **Focus:** GPU kernels and LLM training/inference systems; Attention, MoE routing, quantized GEMM, tensor parallelism, and CUDA Graphs.
- **Languages and credentials:** IELTS 6.5; CET-6; Huawei HCIA-AI. Experienced in English technical documentation and cross-repository source analysis.